
Rincon's Bio Weaver Tool – From Disparate Data to Actionable Analysis

MEDS 2025-2026 Group Project

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Objective

Rincon's biologists are seeking an innovative, data-driven workflow to unify and analyze environmental datasets, particularly special status species occurrence data, used in biological resource evaluations. Students will design an automated system that consolidates these datasets into a single dynamic platform, embedding preliminary analyses within an interactive site that empowers field biologists with real-time, site-specific insights.

Environmental Motivation

California's diverse ecosystems are under mounting pressure from urban development, habitat fragmentation, and accelerating climate change. These forces drive shifts in species distributions and increase the vulnerability of already at-risk plants and wildlife. As environmental stewards, Rincon's biologists conduct biological reviews under the California Environmental Quality Act (CEQA) to evaluate how proposed projects may affect special-status and culturally significant species. However, the complexity of today's environmental landscape demands a more comprehensive and connected approach to understanding species' occurrences. Existing tools and datasets such as the California Department of Fish and Wildlife's California Natural Diversity Database (CNDDDB) and the California Native Plant Society's Rare Plant Inventory, offer vital insights but operate in isolation. Integrating these resources provides an opportunity to reveal ecological relationships that are often overlooked when data remain siloed.

By connecting and visualizing these datasets within a unified analytical framework, Rincon's team can better anticipate where sensitive species may occur, account for environmental drivers such as elevation or habitat type, and promote proactive planning that safeguards biodiversity. This effort represents a step toward more adaptive, data-informed conservation practices that strengthen long-term resource management and ecological resilience across California.

Data Science Need:

Rincon's current approach to compiling species occurrence data is largely manual and fragmented, relying on ad-hoc downloads of CNDDDB and CNPS extracts, manual joins, KMZ requests, and repetitive data formatting steps. This process introduces inefficiencies, duplicate records, and inconsistencies that limit the reliability and reproducibility of analyses. Additionally, reliance on manual exports slows response times and increases the potential for human error.

To address these challenges, Rincon seeks to implement an automated, reproducible data workflow that integrates data from multiple sources into a centralized, cloud-based environment. This workflow will support a dynamic dashboard interface, enabling biologists to interact with cleaned, standardized datasets in real time. The result will be a scalable, efficient system that enhances data integrity, reduces redundancy, and empowers Rincon's biologists to focus on ecological interpretation rather than data wrangling.

Deliverables

1. A reproducible workflow with user tutorials and documentation
2. A GitHub Repository: A private GitHub repository of the tool's design with access to all thoroughly annotated scripts, libraries, and documentation
3. An Interactive Application: A centralized user portal that accepts biologist-defined inputs, automates data analysis, and generates standardized outputs in the form of a dashboard,

application or website. The tool will consolidate occurrence data, visualize species patterns, and deliver ready-to-use results to support Rincon's biological review workflows.

Broader Impacts:

This project presents an exciting and novel opportunity for the Bren School to partner and collaborate with an environmental consulting company in California and MEDs alumni. This new partnership will provide students with exposure to real-world issues environmental consultants face, creating applied learning opportunities and fostering academic-industry collaboration.

Data Access and Availability:

The group will have access to the following datasets to complete the expected deliverables immediately upon project start. Rincon will schedule training within the first two weeks of the project to get students familiar with the public datasets and existing data that Rincon manages. Datasets linked below are publicly available or Rincon-accessible.

Dataset Name	Description	Owned By	Account Required
California Natural Diversity Database (CNDDDB)	Statewide inventory of rare, threatened, endangered or otherwise special-status species and natural communities. Rincon will provide funds for team members to sign up for a 1-year subscription to the database. Students will need to comply with all CNDDDB guidelines when utilizing and presenting data from this source.	California Department of Fish & Wildlife (CDFW)	Yes
California Native Plant Society (CNPS)	The California Native Plant Society (CNPS) Rare Plant Inventory is a statewide database that compiles information on the distribution, ecology, and conservation status of California's rare, threatened, and endangered plant species, serving as a primary reference for environmental review and botanical assessments.	California Native Plant Society (CNPS)	No
US Fish and Wildlife IPaC Tool	The USFWS's Information for Planning and Consultation (IPaC) database is an online tool that provides project-specific information on federally listed species, critical habitats, and other natural resources under USFWS jurisdiction to support environmental review and regulatory compliance.	U.S Fish and Wildlife Service (USFWS)	No
Spotted Owl Observation Data	is a statewide repository maintained by CDFW that houses survey records and location data for both northern and California spotted owls. This data is accessible through the CNDDDB subscription.	California Department of Fish & Wildlife (CDFW)	Yes
GitHub	Rincon will provide students with access to Rincon's GitHub organization as external collaborators.	GitHub	Yes
Shapefiles	Rincon will provide students with example project shapefiles to use as data inputs.	Rincon	No

Project Requirements

Rincon has identified several core requirements for this project, while allowing flexibility for students to determine the most effective methods and technologies to achieve them. The primary objective is to design and implement a user portal that enables biologists to input project-specific parameters, execute automated analyses, and generate standardized outputs.

User inputs may include:

- Uploading a project area shapefile (.shp)
- Selecting a query radius (e.g., 5 miles, 9 miles, etc.)
- Specifying an output file path for saving results

Data processing and outputs may include:

- Automated retrieval and merging of data from multiple biological databases
- Spatial visualization of the processed data within the project's area of interest
- Generation of three CSV files, individual CNDDDB and CNPS search results, plus a merged dataset
- Downloadable tables formatted for inclusion in Rincon's standard biological reports, showing preliminary potential-to-occur findings

Analytical components will integrate spatial and tabular data to assess species occurrences within and around project areas. Analyses may include:

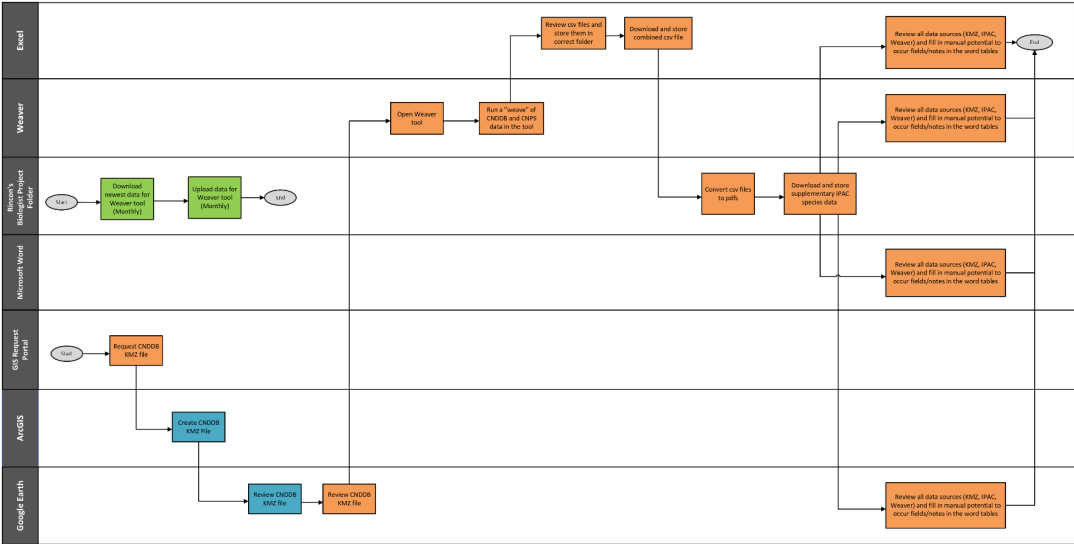
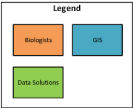
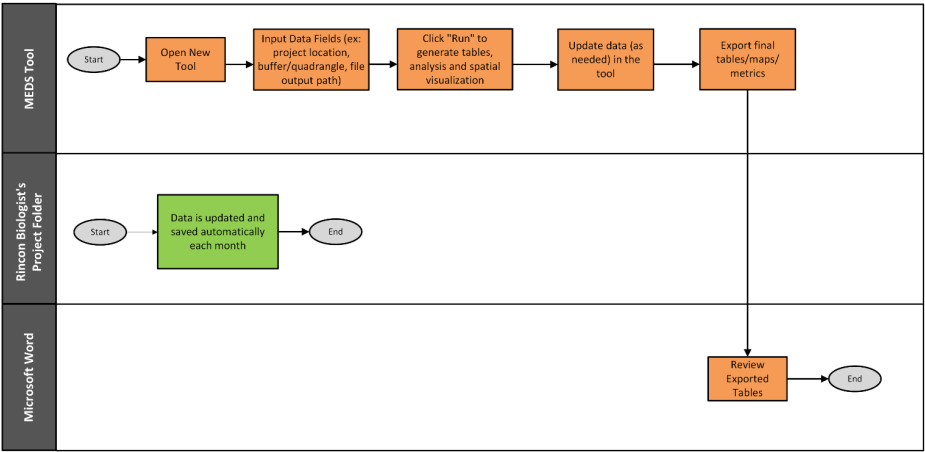
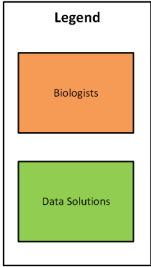
- Estimate potential occurrences and evaluate species presence likelihoods (both presumed extant and potentially extirpated)
- Incorporate data from the USGS 7.5-minute quadrangle covering the project site and adjacent quadrangles, as well as a five-mile buffer
- Account for each species' last observation date, CNDDDB occurrence status (e.g., presumed extant, possibly extirpated, or extirpated), and information from the U.S. Fish and Wildlife Service's IPaC system regarding federally protected species and critical habitats
- Allow users to manually add species not identified in CNDDDB or CNPS search results

Final Deliverables include a transferable, interactive dashboard accessible to Rincon team members, supporting continued use beyond the project.

Technical Requirements may include:

- Building dashboards with spatial visualization elements in a recommended dynamic tool
- Accessing and processing data from Microsoft Excel and KMZ files
- Exporting output tables in a Microsoft Word-compatible format
- Scripting in R, Python, or SQL, based on project design

Rincon's Data Science team has expertise in these technologies and can maintain, and support applications built on the chosen platform. Additionally, Rincon will dedicate time to expose students to ArcGIS and Esri tools, offering hands-on experience with spatial data workflows if the Esri suite of tools is selected for visualization or analysis.



October 16th, 2025

MEDS Capstone Committee
Bren School of Environmental Science & Management
Bren Hall
2400 University of California
Santa Barbara, California 93117

Subject: Rincon Letter of Support for the Bren MEDS Program Group Project Committee

To Whom it May Concern,

This letter reaffirms Rincon Consultants, Inc.'s support for the MEDS Capstone Group Project Proposal. Rincon is committed to providing all data and resources necessary for the successful completion of this project. Most of the required datasets are publicly available and exportable, and any Rincon-specific data or materials will be provided at the start of the project. All data will be shared without the requirement of a non-disclosure agreement.

We view this collaboration as a unique opportunity for students to apply practical data science skills to a real-world environmental challenge aligned with California's biological reporting requirements. Rincon will benefit significantly from the development of a clear, reproducible workflow that enables efficient access and analysis of the data needed to complete biological resource evaluations effectively.

To support this effort, Rincon will allocate \$600 to fund a one-year subscription to the CDFW/CNDDDB database, licensed independently through Bren, allowing students to establish an account and obtain necessary data for the project.

We also recognize the need for a private GitHub repository, consistent with past projects, and reaffirm our shared goal of promoting open collaboration and transparency. Public sharing will be permitted for select visualizations, screen recordings, and documentation that exclude written code. Ownership of the repository and all associated intellectual property will be transferred to the client upon project completion. Access to the private repository will be restricted to project collaborators.

Rincon is excited to collaborate with the student team on this project and looks forward to working alongside them to explore opportunities for enhancing Rincon's existing tools and data capabilities.



Rincon Consultants, Inc.

180 North Ashwood Avenue
Ventura, California 93003
805-644-4455

Thank you for your time and consideration.

Sincerely,

Rincon Consultants, Inc

A handwritten signature in blue ink, appearing to read 'C. Huff'.

Craig Huff

Chief Information and Security Officer

A handwritten signature in blue ink, appearing to read 'Robin Murray'.

Robin Murray

Senior Supervising Biologist